NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-6106/40B(USAF)
22 June 1993
SUPERSEDING
MIL-R-6106/40A(USAF)
21 March 1985

PERFORMANCE SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, TYPE I, MAGNETIC LATCH, PERMANENT MAGNET DRIVE, LOW LEVEL TO 10 AMPERES, 3 PDT, HERMETICALLY SEALED

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-6106.

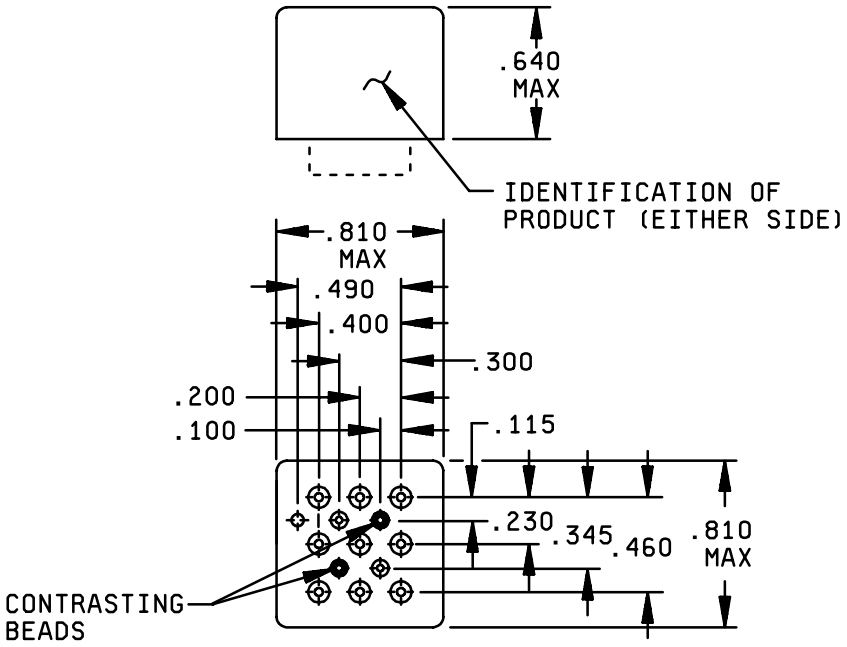
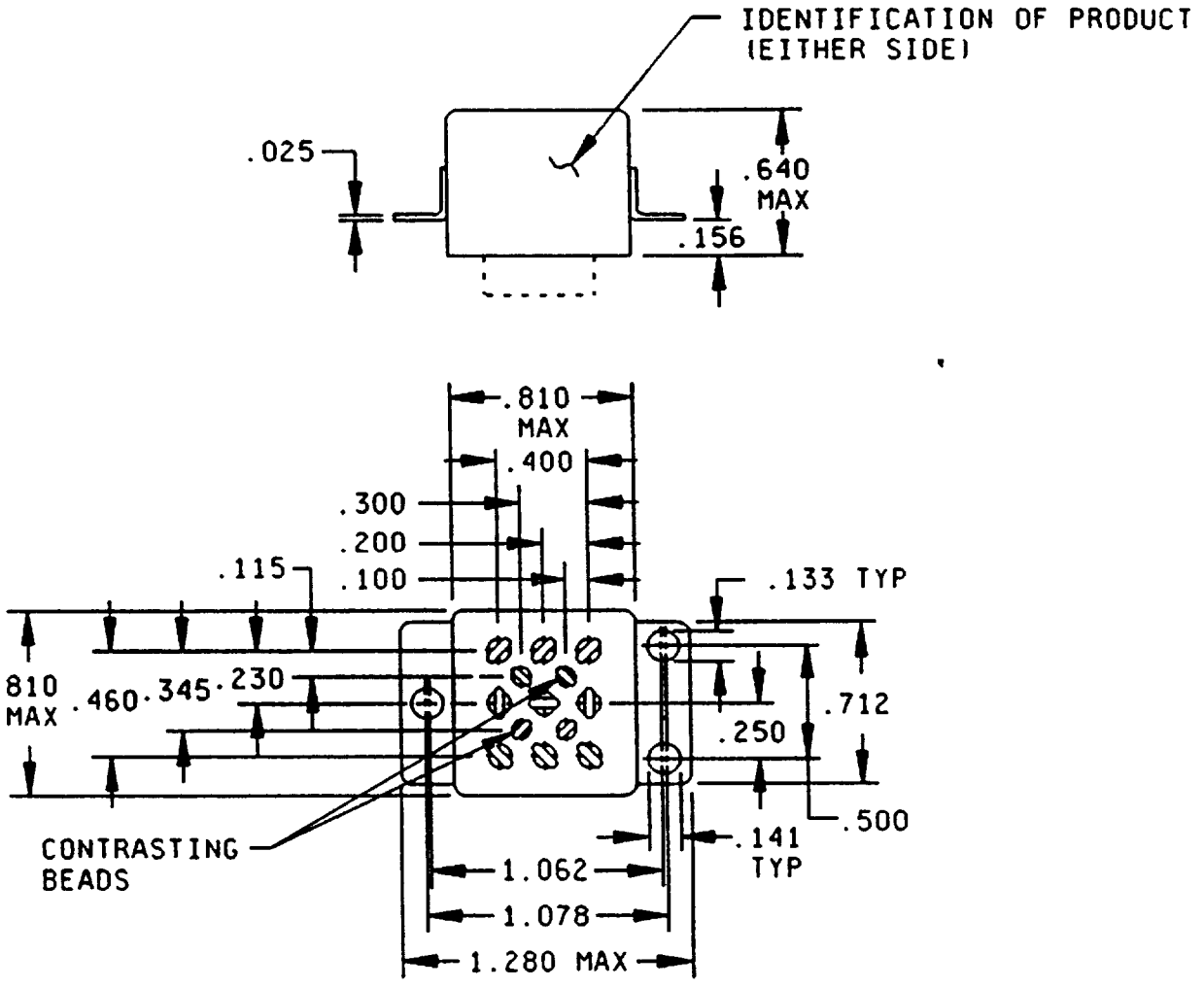


FIGURE 1. Outline drawing.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

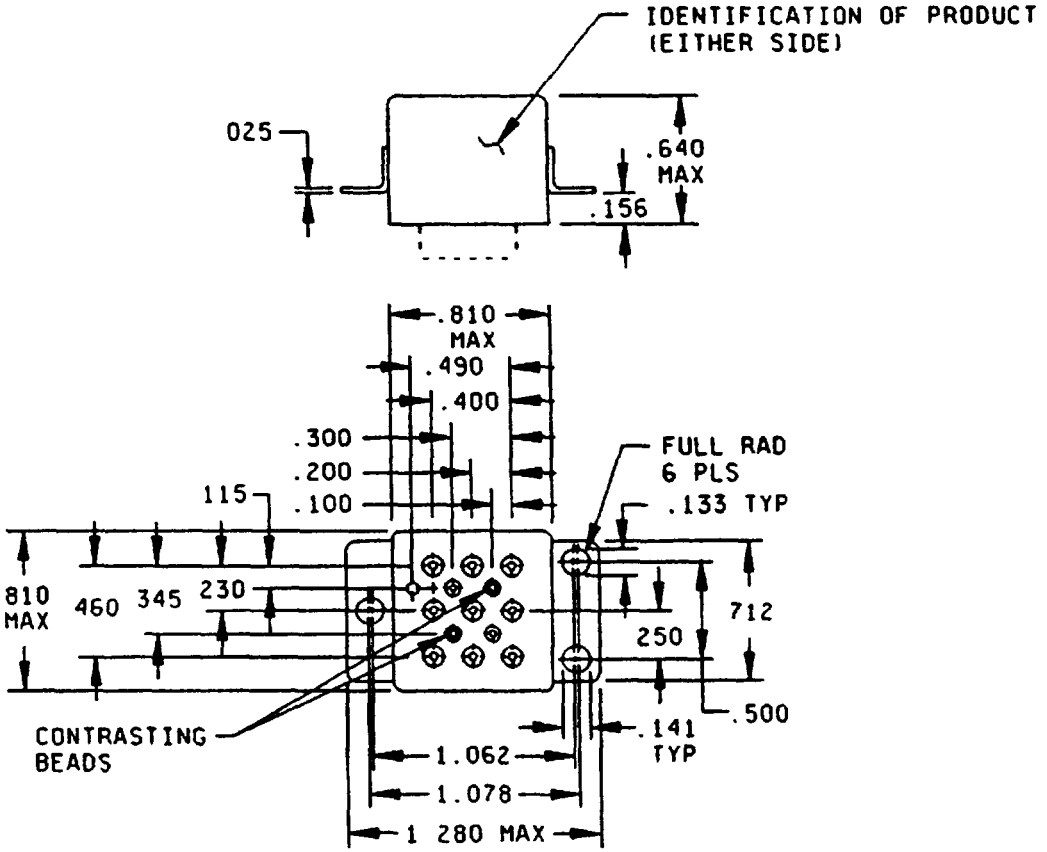
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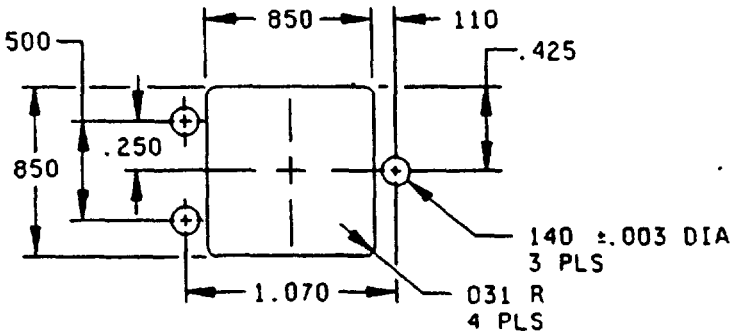
-001 (SOLDER HOOK)
CONFIGURATION B

FIGURE 1 Outline drawing - Continued

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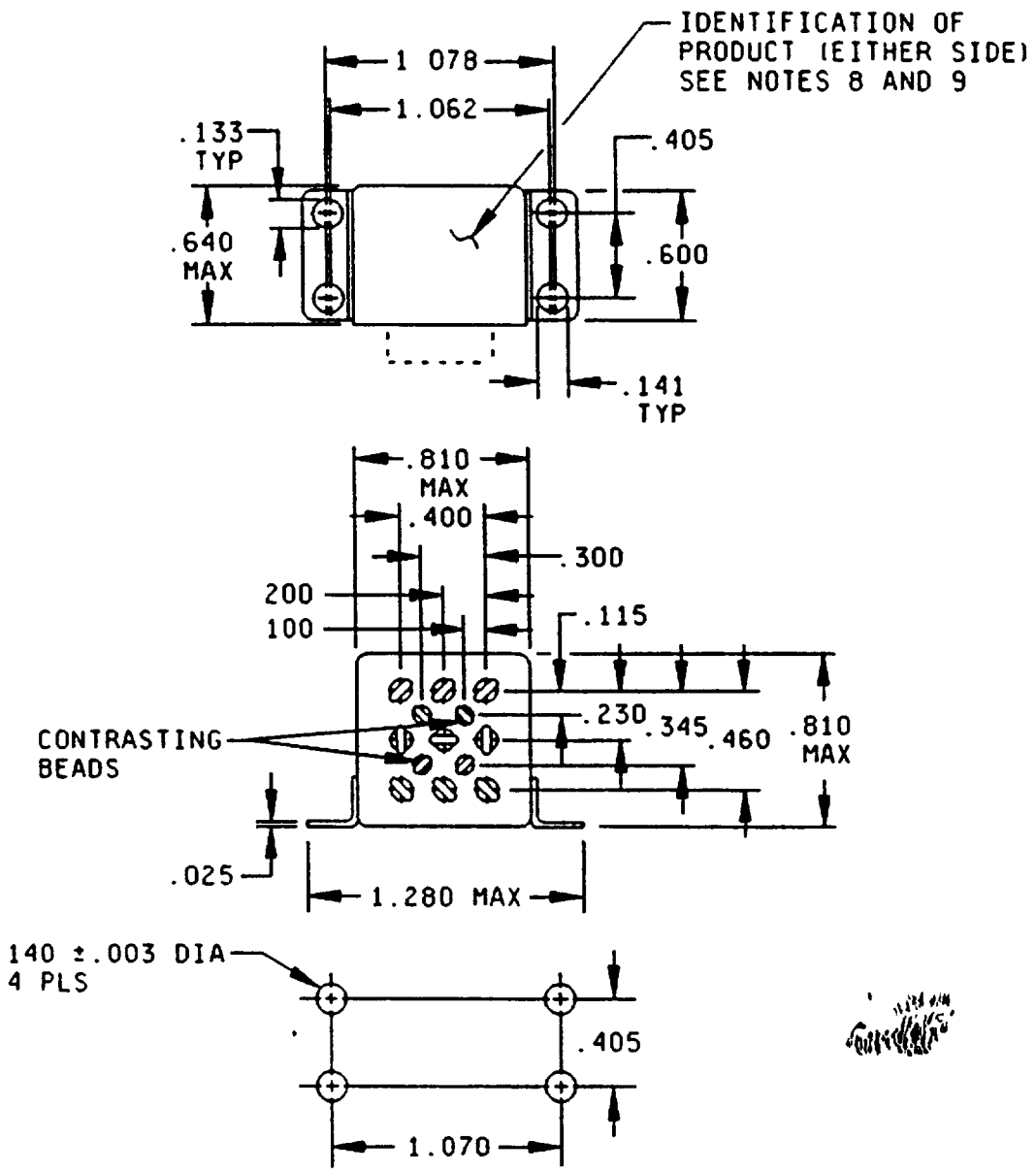
-002 (SOCKET PIN) AND -003 (SOLDER PIN)
CONFIGURATION B (CONT'D)



MOUNTING HOLE LAYOUT (-001, -002, -003)

FIGURE 1 Outline drawing - Continued

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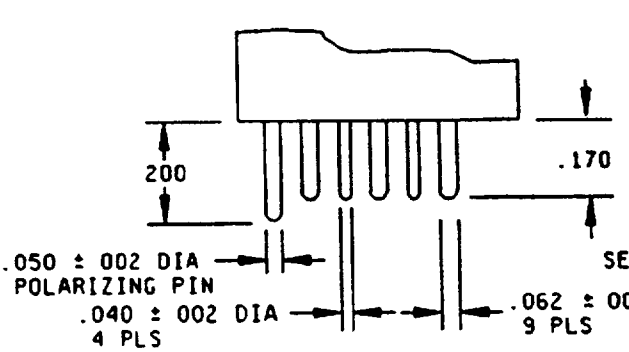


MOUNTING HOLE LAYOUT

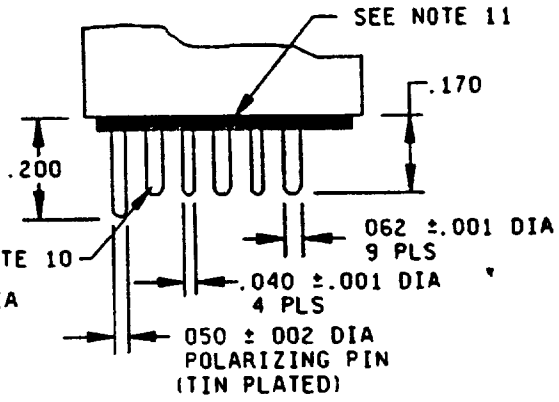
-004 AND -006 (SOLDER HOOK)
CONFIGURATION C

FIGURE 1 Outline drawing - Continued

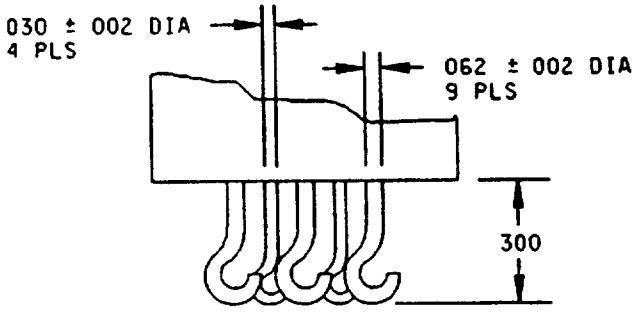
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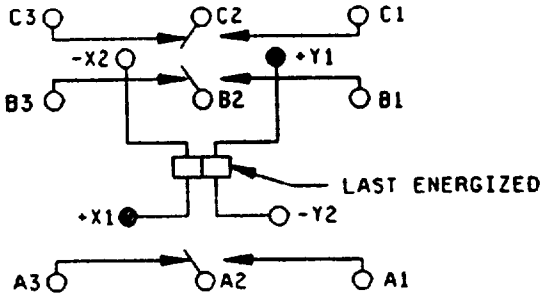
SOLDER PIN FOR -003, -005, -006
(ALL PINS TIN PLATED)



SOCKET PIN FOR -002



SOLDER HOOK FOR -001, -004
(ALL HOOKS TIN PLATED)



CIRCUIT DIAGRAM

TO CLOSE NO 1 CONTACTS (OPERATE)
ENERGIZE X1 AND X2
TO CLOSE NO 3 CONTACTS (RESET)
ENERGIZE Y1 AND Y2

FIGURE 1 Outline drawing - Continued

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
001	0 03	062	1 57	.200	5.08	640	16.26
002	0 05	.100	2.54	230	5 84	712	18 08
003	0 08	110	2 79	.250	6 35	810	20 57
005	0 13	115	2 92	300	7 62	850	21.59
025	0.64	.133	3.38	.345	8 76	1 062	26.97
030	0.76	.140	3.56	.400	10 16	1 070	27.17
031	0 79	141	3.58	.425	10.80	1.078	27.38
040	1 02	.156	3 96	460	11.68	1 280	32.51
050	1.27	.170	4 32	.500	12.70		

NOTES

1. Dimensions are in inches
2. Metric equivalents are given for general information only
3. Unless otherwise specified, tolerance is ± 010 inch (25 4 mm)
4. There shall be affix to the relay a legible circuit diagram that identifies each terminal location specified
5. These relays are polarized monostable
6. This relay shall not operate or be damaged by reverse polarity Semiconductors shall not be used for this purpose
7. Permanent magnet drive consists of a permanent magnet with its flux path switched and combined with the electromagnetic flux
8. Applicable to configuration C only The circuit diagram, manufacturer's PIN, and the military PIN shall be marked on the near side The remaining portion of the nameplate data shall be marked on the far side
9. Applicable to configuration C only Relays shall be marked with the manufacturer's name or source code and date code In addition, ER relays shall be marked with the serial number. Marking shall be with the bottom of the print adjacent to the near side
10. Socket pin terminals, except the polarizing pin, shall be gold-plated in accordance with MIL-G-45204, type II, class I The nickel underplating shall be 50 to 150 microinches thick minimum
11. Silicone rubber gasket, AMS 3332, shore hardness 25 ±5, thickness .050 ± 005 inch (1 27 ±0 13 mm)
12. Back EMF (transient voltage): 42 V dc maximum
13. JANTX or equivalent screened semiconductors shall be used in ER relays with internal coil suppression Relays using suppression devices shall continue to operate should the suppression circuit be in a failure mode Diodes shall have a peak inverse voltage of 600 V dc minimum

FIGURE 1 Outline drawing - Continued

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REQUIREMENTS

CONTACT DATA

Life Unless otherwise specified 50,000 cycles

Load ratings

High level (relay case grounded)

Resistive 10 amperes at 28 V dc, 115 V ac, 400 Hz, 1 phase, 115/200 V ac, 400 Hz, 3 phase

Inductive 6 amperes at 28 V dc (life· 10,000 cycles), 8 amperes at 115 V ac, 400 Hz, 1 phase, 115 V ac, 400 Hz, 3 phase (life 20,000 cycles)

Motor: 4 amperes at 28 V dc, 115 V ac, 400 Hz, 1 phase; and 115/200 V ac, 400 Hz, 3 phase.

Lamp: 2 amperes at 28 V dc and 115 V ac, 400 Hz, 1 phase.

Low level: 10 to 50 µA at 10 to 50 mV dc or peak ac 1/ 2/ 3/

Intermediate current· Applicable

Transfer load Not applicable.

Mechanical life (reduced current) 1.25 amperes at 28 V dc, 115 V ac, 1 phase and 3 phase (life 200,000 cycles)

Mixed loads. Applicable 4/

Contact voltage drop or resistance 1/ 2/ 3/

High level
Initial 0 150 volt maximum
After life 0 175 volt maximum

Low level
Initial 0 050 ohm maximum
After life 0 150 ohm maximum

Contact bounce 1 0 millisecond maximum

Break bounce (normally open contacts only): 0 1 millisecond maximum (applicable to dc suppressed coils only)

1/ For low level, the following shall apply:
a During endurance and operational reliability testing, contact load shall be 10 to 50 microamperes, 10 to 50 millivolts open circuit voltage, 100 ohms maximum contact resistance
b Static contact resistance shall be performed at 50 mA maximum, 50 mV maximum

2/ For all ER relays, the alternate low level test of operational reliability shall be used

3/ For group A contact voltage drop test, high level testing shall be performed first, followed by low level testing The contacts shall not make or break the high level load

4/ Relay shall be capable of switching low level while switching any of the other rated loads on adjacent contacts

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Overload current 30 amperes dc
60 amperes ac

Rupture current 40 amperes dc
80 amperes ac

Time current relay characteristics Applicable (see table I)

TABLE I Time current relay characteristics 1/ 2/

1	20 A	1 hour
2	30 A	20 seconds
3	75 A	2 seconds
4	150 A	.53 second

1/ CAUTION: Compare with time current characteristics of the associated circuit protective device

2/ All relays shall withstand overload and fault currents. Relays must be able to sustain five applications (make and carry only) of power concurrently on adjacent poles at each of four different current levels in the sequence listed in table I. Separate relays shall be tested at 28 V dc and 115/200 V ac, 400 Hz, 3 phase. Cooling time between successive applications shall be 30 minutes Tests shall be performed on both normally open and normally closed contacts. There shall be no failures or evidence of welding or sticking and relays shall pass contact voltage drop at the conclusion.

COIL DATA 5/ (See table II)

Operate time 6 milliseconds maximum with rated coil voltage.

Release time 6 milliseconds maximum from rated coil voltage

Duty rating: Continuous

ELECTRICAL DATA

Insulation resistance, initial 100 megohms

After life or environmental tests 50 megohms

Dielectric withstanding voltage (sea level)	Initial V rms (60 Hz)	After life tests V rms (60 Hz)
Coil to case	500	350
Coil to coil	500	350
All other points	1,250	1,000

Dielectric withstanding voltage (altitude) 6/	80,000 feet V rms (60 Hz)	300,000 feet V rms (60 Hz)
Coil to case	250	500
Coil to coil	250	500
All other points	350	500

5/ CAUTION Due to possible interaction of relay magnetic fields, the following spacing requirements, as a minimum, shall be considered in dense packaging situations:
a Row to row assisting fields 125 inch
b Row to row opposing fields 1875 inch
c Side to side alternating fields 0625 inch
d Side to side like fields 125 inch

6/ Dielectric may be improved by suitable insulation of terminals and wiring after installation

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TABLE II. Dash number and characteristics

Military PIN M6106/40-	Coil data									Terminal type	Configuration	Mating socket
	Coil	Rated			Max		Max pick-up voltage					
		Volts 1/	Freq Hz	Res ±10% at 25°C	Volts 2/	Amp at 25°C	Normal 3/	High test temp	Cont current test			
001	X1,X2 Y1,Y2	28	DC	600	29	.054	18	19.8	22.5	Solder hook	B	N/A
002	"	"	"	"	"	"	"	"	"	Socket pin	B	MIL-S-12883/46-002
003	"	"	"	"	"	"	"	"	"	Solder pin	B	N/A
004	"	"	"	"	"	"	"	"	"	Solder hook	C	"
005	"	"	"	"	"	"	"	"	"	Solder pin	A	"
006	"	"	"	"	"	"	"	"	"	Solder pin	C	"

1/ CAUTION The use of any coil voltage less than the rated voltage will compromise the operation of the relay

2/ When maximum ambient temperature does not exceed +85°C, the maximum coil voltage shall be 32 V dc

3/ Over the temperature range

ENVIRONMENTAL CHARACTERISTICS

Operating temperature range -70°C to +125°C

Maximum altitude rating 300,000 feet

Acceleration Applicable (15 g's maximum)

Shock (specified pulse) MIL-STD-202, method 213, test condition C (100 g's), except configurations A and B peak g value shall be 200 g's Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum for open contacts

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Vibration (sinusoidal) Configurations A and B shall be 30 g's, 70 to 3,000 Hz Configuration C shall be 20 g's, 57 to 3,000 Hz Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum for open contacts.

Vibration (random). MIL-STD-202, method 214, for configurations A and B, test condition IG (0 4 g^2/Hz), for configuration C, test condition IE (0.2g^2/Hz). Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum for open contacts

PHYSICAL DATA

Terminal strength

Solder hook terminals·

Pull force (0.030 diameter) 3 ±0 3 pounds.

Pull force (0.062 diameter)· 10 ±1 pounds.

Solder pin terminals.

Bend test Applicable to coil terminals only

Pull force (0 040 diameter). 5 ±0.5 pounds.

Pull force (0 062 diameter) 10 ±1 pounds

Plug-in terminals

Bend test. Applicable to coil terminals only

Pull force (0 040 diameter) 5 ±0.5 pounds

Pull force (0 062 diameter) 10 ±1 pounds.

Terminal solderability Applicable to solder pin and solder hook terminals only.

Dimensions and configurations See figure 1

Weight (maximum) 0 068 pound (30 87 grams)

Seal Hermetic, relays shall be sealed by welding i.e , laser, TIG (tungsten inert gas) or other suitable means as approved by the qualifying activity

Construction (internal and external). All welded, except that coil magnet wire to coil lead wire shall be soldered

Identification of product Applicable

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Qualification similarity See table III

TABLE III Qualification by similarity

PIN M6106/40-	Basic qualification A (DC)	Dynamics Socket/pins A 1/	B	C	D	Environmental A (DC)
001	4		Config B 2/			4
002	4	Config B 2/				4 2/
003	4		Config B 2/			4
004	4			Config A 2/		4
005	4				Config A 2/	4
006	4			Config A 2/		4

1/ Terminals Socket pin type terminals are considered worst case. Solder pins and hooks may be qualified by similarity

2/ Socket pins (with gasket seals), are considered nonsimilar to solder pins and solder hooks Each type must be tested

Revision letters are not used on this revision to denote changes due to the extensiveness of the changes

CONCLUDING MATERIAL

Custodian
Air Force - 85

Review activities
Air Force - 99
DLA - ES

Preparing activity
Air Force - 85

Agent
DLA - ES

(Project 5945-F736-08)

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